for loop

A for loop is used to repeat a block of code for each item in a sequence, such as a list, string, tuple, or range.

simple example

In [10]:

```
for i in range(5):
    print(i)
```

```
0
1
2
3
4
```

1.print 1 to 5

In [13]:

```
for i in range(1,6): # 1=start , 6=end
    print(i)
```

```
1
2
3
4
5
```

2.Loop through a string

In [18]:

```
a="Python"
for i in a:
    print(i,end=" ")
```

```
P y t h o n
```

In [20]:

```
a="Python"
for i in a:
    print(i)
```

```
P
y
t
h
o
n
```

3.Loop through a list

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In [23]:

```
names = ["Reeta","Manoj","Raja","Selvi"]
for i in names:
    print(i)
```

```
Reeta
Manoj
Raja
Selvi
```

4.range

In [26]:

```
for i in range(10):
    print(i)
```

```
0
1
2
3
4
5
6
7
8
9
```

In [28]:

```
num=1
for i in range(1,11):
    print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

In [30]:

```
num=1
for i in range(1,11,2):
    print(i)
```

```
1
3
5
7
9
```

5.break

The break statement is used to stop a loop immediately, even if the loop has not finished.

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In [51]:

```
names = ["Reeta","Manoj","Raja","Selvi"]
for i in names:
    if i=="Raja":
        break
    print(i)
```

```
Reeta
Manoj
```

6.continue

The continue statement skips the current iteration and moves to the next iteration of the loop.

In [54]:

```
names = ["Reeta","Manoj","Raja","Selvi"]
for i in names:
    if i=="Manoj":
        continue
    print(i)
```

```
Reeta
Raja
Selvi
```

7.pass

pass is used when you want to leave a block of code empty temporarily without getting an error.

In [72]:

```
age = 20
if age >= 18:
    pass
else:
    print("Not eligible")
```

8.nested loop

In [58]:

```
for i in range(3):
    for j in range(2):
        print(i, j)
```

```
0 0
0 1
1 0
1 1
2 0
2 1
```

In [60]:

```
a=[1,2,3]
b=[10,20,30]
for i in a:
```

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```
    for j in b:
        print(i,j)
```

```
1 10
1 20
1 30
2 10
2 20
2 30
3 10
3 20
3 30
```

Practice Question

1.Write a Python program to print numbers from 1 to 10 using a for loop.

In [76]:

```
num=1
for i in range(num,11):
    print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

2.Write a Python program to print all even numbers from 1 to 50.

In [81]:

```
for i in range(1,51):
    if i%2==0:
        print(i,end=" ")
```

```
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50
```

3.Write a Python program to calculate the sum of numbers from 1 to n.

In [96]:

```
sum=0
for i in range(1,5+1):
    sum +=i
print(sum)
```

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```
15
```

In [98]:

```
sum=0
for i in range(1,5+1):
    sum +=i
    print(sum)
```

```
1
3
6
10
15
```

4.Write a Python program to calculate the factorial of a number using a for loop.

In [111]:

```
num=5
fact =1
for i in range(1,num+1):
    fact *=i
print(fact)
```

```
120
```

5.Write a Python program to print the multiplication table of a number entered by the user.

In [124]:

```
for i in range(1,10+1):
    print(2,"x",i,"=",2*i)
```

```
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20
```

6.Write a Python program to reverse a string using a for loop.

In [137]:

```
name="Python"
a=""
for i in name:
    a=i+a
print(a)
```

```
nohtyP
```

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7.Write a Python program to count the vowels in a string using a for loop.

In [154]:

```
name="Reeta"
count=0
for i in name.lower():
    if i=="a" or i=="e" or i=="i" or i=="o" or i=="u":
        count +=1
print(count)
```

```
3
```

8.Write a Python program to find the largest number in a list without using max().

In [163]:

```
num=[1,4,3,6,8]
l=num[0]
for i in num:
    if l <i:
        l=i
print(l)
```

```
8
```

9.Write a Python program to print numbers from 1 to 20, skipping numbers divisible by 3 using continue.

In [168]:

```
for i in range(1,21):
    if i%3==0:
        continue
    print(i)
```

```
1
2
4
5
7
8
10
11
13
14
16
17
19
20
```

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10.Write a Python program using a for loop to print each character in the word "Python".

In [172]:

```
a="Python"
for i in a:
    print(i)
```

```
P
y
t
h
o
n
```

task

In [3]:

```
for i in range(1,31):
    if i>1:
        if i%2==0 and i!=2:
            print(i,"np")
        else:
            print(i,"p")
    else:
        print(i,"np")
```

```
1 np
2 p
3 p
4 np
5 p
6 np
7 p
8 np
9 p
10 np
11 p
12 np
13 p
14 np
15 p
16 np
17 p
18 np
19 p
20 np
21 p
22 np
23 p
24 np
25 p
```

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```
26 np
27 p
28 np
29 p
30 np
```

In [1]:

```
# max and mim
l=[2,5,4,3,7]
m=l[0]
mi=l[0]
for i in l:
    if i>m:
        m=i
    elif i<mi:
        mi=i
print(m)
print(mi)
```

```
7 7
7 2
```

In [7]:

```
l=[1,2,5,21,5]
print(len(l))
```

```
5
```

find len without inbuilt

In [22]:

```
l=[1,2,5,21,5]
count=0
for i in l:
    count = count+1
print(count)
```

```
5
```

In [ ]:

```
0=0+1  -1
1=1+1  -2
2=2+1  -3
3=3+1  -4
4=4+1  -5
```

even number only i want print in list

In [30]:

```
l=[5,6,8,2,1,13,15,17]
#even=[]
for i in l:
    if i%2==0:
        print(i)
```

```
6
8
```

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2

In [40]:

```
l=[5,6,8,2,1,13,15,17]
for i in l:
    if i%2==0:
        print("[",i,"]",end=" ")
```

```
[ 6 ] [ 8 ] [ 2 ]
```

In [51]:

```
l=[5,6,8,2,1,13,15,17]
e=[]
for i in l:
    if i%2==0:
        e.append(i)
print(e)
```

```
[6, 8, 2]
```

remove duplicate

In [15]:

```
l = [2, 3, 3, 5, 6, 1, 2, 1]
d = []
for i in l:
    if i not in d:
        d.append(i)
print(d)
```

```
[2, 3, 5, 6, 1]
```

In [55]:

```
l=[2,3,3,5,6,1,2,1]
d=[]
for i in l:
    if i==i:
        print(i)
```

```
2
3
3
5
6
1
2
1
```

In [ ]:

```
# while loop
```

In [ ]:

```
a=[]
while len(a)<5:
    l=int(input("enter l:"))
    a.append(l)
print(a)
```

In [ ]:

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```
max,min
count
duplicate
even
while(len)
```

In [9]:

```
#palindrom
num=121
og=num
rev=0
while num >0:
    digit = num % 10
    rev = rev *10+digit
    num //=10
if og == rev:
    print("palindrome")
else:
    print("not palindrome")
```

```
palindrome
```

In [13]:

```
l = [2,2,3,4,3,4,2,6,7,8]
a = {}
for i in l:
    if i in a:
        a[i] += 1
    else:
        a[i] = 1
print(a)
```

```
{2: 3, 3: 2, 4: 2, 6: 1, 7: 1, 8: 1}
```

In [ ]:

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